King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 102 Major Exam II Term 253

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The area of the surface generated by revolving the curve $y = 1 - \dfrac{x^2}{4}$ on the interval $[0, 2]$ about the y-axis is equal to

(a) $\dfrac{4\pi}{3}\left(2\sqrt{2} - 1\right)$

(b) $\dfrac{8\pi}{3}\left(2\sqrt{2} + 1\right)$

(c) $\dfrac{8\pi}{3}\left(2\sqrt{2} - 1\right)$

(d) $\dfrac{4\pi}{3}\left(2\sqrt{2} + 1\right)$

(e) $\dfrac{4\pi}{3}\left(\sqrt{2} + 1\right)$

2.

$$\int_{\frac{1}{2\sqrt{3}}}^{\frac{1}{2}} \frac{dx}{x^2\sqrt{4x^2+1}} =$$

(a) $4 - 2\sqrt{2}$

(b) $2 + \sqrt{2}$

(c) $2 - 4\sqrt{2}$

(d) $2 + 4\sqrt{2}$

(e) $4 + 2\sqrt{2}$

3.

$$\int \frac{4}{(x-1)(x^2+1)}\,dx =$$

(a) $\ln\dfrac{(x-1)^2}{x^2+1} + 2\arctan x + C$

(b) $\ln\dfrac{x-1}{(x^2+1)^2} - 2\arctan x + C$

(c) $\ln\dfrac{x-1}{(x^2+1)^2} + \operatorname{arccot} x + C$

(d) $\ln\dfrac{(x-1)^2}{x^2+1} - 2\arctan x + C$

(e) $\ln\dfrac{(x-1)^2}{x^2+1} - \arctan x + C$

4.

$$\int \cos^3 x \, \sin^2 x \, dx =$$

(a) $-\dfrac{\sin^3 x}{3} - \dfrac{\sin^5 x}{5} + C$

(b) $\dfrac{\sin^3 x}{3} + \dfrac{\cos^5 x}{5} + C$

(c) $\dfrac{\sin^3 x}{3} + \dfrac{\sin^5 x}{5} + C$

(d) $\dfrac{\cos^3 x}{3} - \dfrac{\cos^5 x}{5} + C$

(e) $\dfrac{\sin^3 x}{3} - \dfrac{\sin^5 x}{5} + C$

5. Using the substitution $t = \tan \dfrac{x}{2}$,

$$\int_{\pi/3}^{\pi/2} \frac{1}{1 + \sin x - \cos x} \, dx =$$

(a) $\ln \dfrac{\sqrt{2} - 1}{3}$

(b) $\ln \dfrac{\sqrt{2} + 1}{3}$

(c) $\ln \dfrac{\sqrt{3} + 1}{2}$

(d) $\ln \dfrac{\sqrt{3} - 1}{2}$

(e) $\ln \dfrac{\sqrt{3}}{2}$

6. The volume of the solid generated by revolving the region bounded by the graphs of $y = \sqrt{25 - x^2}$, $y = 0$, $x = 0$, and $x = 3$ about the x-axis, is equal to

(a) 44π

(b) 11π

(c) 77π

(d) 22π

(e) 66π

7.

$$\int_2^{7/2} \sqrt{5+4x-x^2}\,dx =$$

(a) $\dfrac{\pi}{4}+\dfrac{9\sqrt{3}}{8}$

(b) $\dfrac{\pi}{2}+\dfrac{\sqrt{3}}{8}$

(c) $\dfrac{3\pi}{4}+\dfrac{9\sqrt{3}}{8}$

(d) $\dfrac{3\pi}{4}+\dfrac{\sqrt{3}}{8}$

(e) $\dfrac{\pi}{4}+\dfrac{\sqrt{3}}{8}$

8. If $\dfrac{-13+9x-x^2}{(x-2)(x-1)^2}=\dfrac{A}{x-2}+\dfrac{B}{x-1}+\dfrac{C}{(x-1)^2}$, then $A+B+C=$

(a) -3

(b) 4

(c) 0

(d) 3

(e) 5

9.

$$\int_1^2 \frac{(\ln x)^2}{x^2}\,dx =$$

(a) $1-\ln 2+\dfrac{(\ln 2)^2}{2}$

(b) $1-\ln 2-\dfrac{(\ln 2)^2}{2}$

(c) $1+\ln 2-\dfrac{(\ln 2)^2}{2}$

(d) $1+\dfrac{(\ln 2)^2}{2}$

(e) $1+\ln 2+\dfrac{(\ln 2)^2}{2}$

10.

$$\int_0^{\pi} x \sin(2x)\, dx =$$

(a) $-\dfrac{\pi}{4}$

(b) $-\dfrac{\pi}{2}$

(c) π

(d) $\dfrac{\pi}{2}$

(e) $\dfrac{\pi}{4}$

11.

$$\int \frac{4-2t}{3t^2-12t-1}\, dt =$$

(a) $-\ln\left|\sqrt[3]{3t^2-12t-1}\right| + C$

(b) $\ln\left|\sqrt{3t^2-12t-1}\right| + C$

(c) $\ln\left|\sqrt{3t^2+12t+1}\right| + C$

(d) $-\ln\left|\sqrt{3t^2-12t-1}\right| + C$

(e) $\ln\left|\sqrt[3]{3t^2-12t-1}\right| + C$

12.

$$\int_0^{\pi/4} \tan^4 x\, dx =$$

(a) $\dfrac{\pi}{4} - \dfrac{2}{3}$

(b) $\dfrac{\pi}{4} + \dfrac{4}{3}$

(c) $\dfrac{\pi}{4} + \dfrac{2}{3}$

(d) $\dfrac{\pi}{2} + \dfrac{4}{3}$

(e) $\dfrac{\pi}{4} - \dfrac{4}{3}$

13. The arc length of the graph of the function $f(x) = \ln(\sec x)$ over the interval $\left[0, \frac{\pi}{4}\right]$ is equal to

(a) $\ln(\sqrt{3} - \sqrt{2})$

(b) $\ln(\sqrt{3} + 1)$

(c) $\ln(\sqrt{3} - 1)$

(d) $\ln(\sqrt{2} - 1)$

(e) $\ln(\sqrt{2} + 1)$

14. The volume of the solid generated by revolving the region bounded by the graphs of $y = x^2$ and $y = 2 - x^2$ about the line $x = 1$, is equal to

(a) $\frac{14\pi}{3}$

(b) π

(c) $\frac{16\pi}{3}$

(d) 3π

(e) $\frac{4\pi}{3}$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	$S = 2\pi \int_0^2 x\sqrt{1+x^2/4}\,dx = \frac{8\pi}{3}(2\sqrt{2}-1)$.
2	**(a)**	$2x = \tan\theta$: $\int_{\pi/6}^{\pi/4} 2\csc\theta\cot\theta\,d\theta = 4-2\sqrt{2}$.
3	**(d)**	$\frac{2}{x-1} - \frac{2x+2}{x^2+1}$, so $\ln\frac{(x-1)^2}{x^2+1} - 2\arctan x + C$.
4	**(e)**	$u = \sin x$: $\int(u^2-u^4)du = \frac{\sin^3 x}{3} - \frac{\sin^5 x}{5} + C$.
5	**(c)**	$\int \frac{dt}{t(t+1)}$ from $\frac{1}{\sqrt{3}}$ to $1 = \ln\frac{\sqrt{3}+1}{2}$.
6	**(e)**	Disks: $\pi\int_0^3(25-x^2)dx = \pi(75-9) = 66\pi$.
7	**(c)**	$9-(x-2)^2$, $u = 3\sin\theta$: $\frac{9}{2}\left(\frac{\pi}{6}+\frac{\sqrt{3}}{4}\right) = \frac{3\pi}{4}+\frac{9\sqrt{3}}{8}$.
8	**(b)**	$A=1$ $(x=2)$, $C=5$ $(x=1)$, $A+B=-1 \Rightarrow B=-2$; sum $=4$.
9	**(b)**	Parts twice: $F = -\frac{(\ln x)^2+2\ln x+2}{x}$, so the value is $1-\ln 2-\frac{(\ln 2)^2}{2}$.
10	**(b)**	By parts: $\left[-\frac{x\cos 2x}{2}\right]_0^{\pi} + 0 = -\frac{\pi}{2}$.
11	**(a)**	$4-2t = -\frac{1}{3}(6t-12)$, so the integral is $-\frac{1}{3}\ln\lvert 3t^2-12t-1\rvert + C$.
12	**(a)**	$\tan^4 x = \tan^2 x\sec^2 x - \sec^2 x + 1$: $\left[\frac{\tan^3 x}{3} - \tan x + x\right]_0^{\pi/4} = \frac{\pi}{4}-\frac{2}{3}$.
13	**(e)**	$f' = \tan x$: $L = \int_0^{\pi/4}\sec x\,dx = \ln(\sqrt{2}+1)$.
14	**(c)**	Shells: $2\pi\int_{-1}^{1}(1-x)(2-2x^2)dx = \frac{16\pi}{3}$.